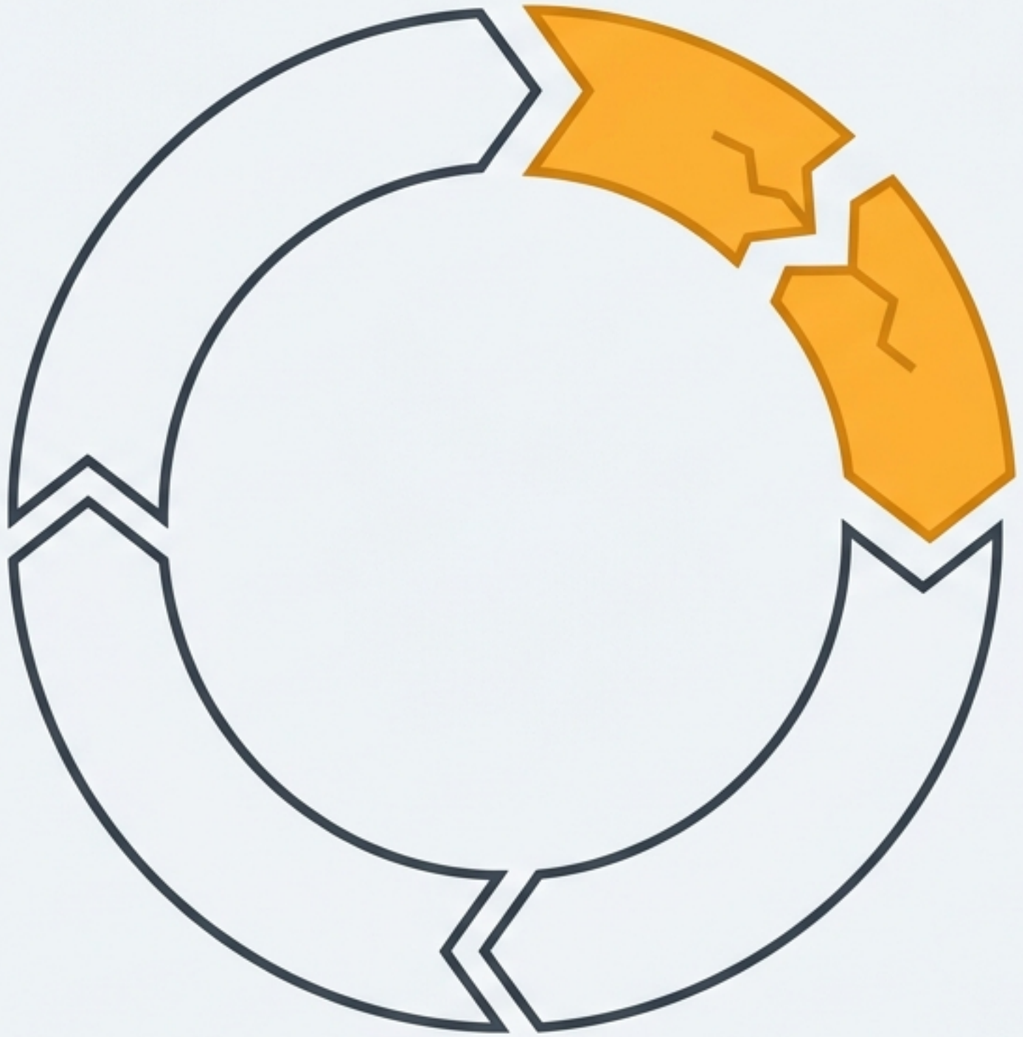
Auto Scaling Didn’t Heal Itself

Lab 08 · Incident Response & Post-Mortem Runbook



AWS Day 2 Operations & Troubleshooting · NorthBank

NotebookLM

[SEV-2]

Priya
(Team Lead)

03:15 AM

We lost a **web instance overnight** and **Auto Scaling** never replaced it.



Impact: Reduced application availability, potential latency increase.
Related Services: AWS Auto Scaling, EC2



Expected Capacity:
2

Current Capacity:
1

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Terminal Timeline

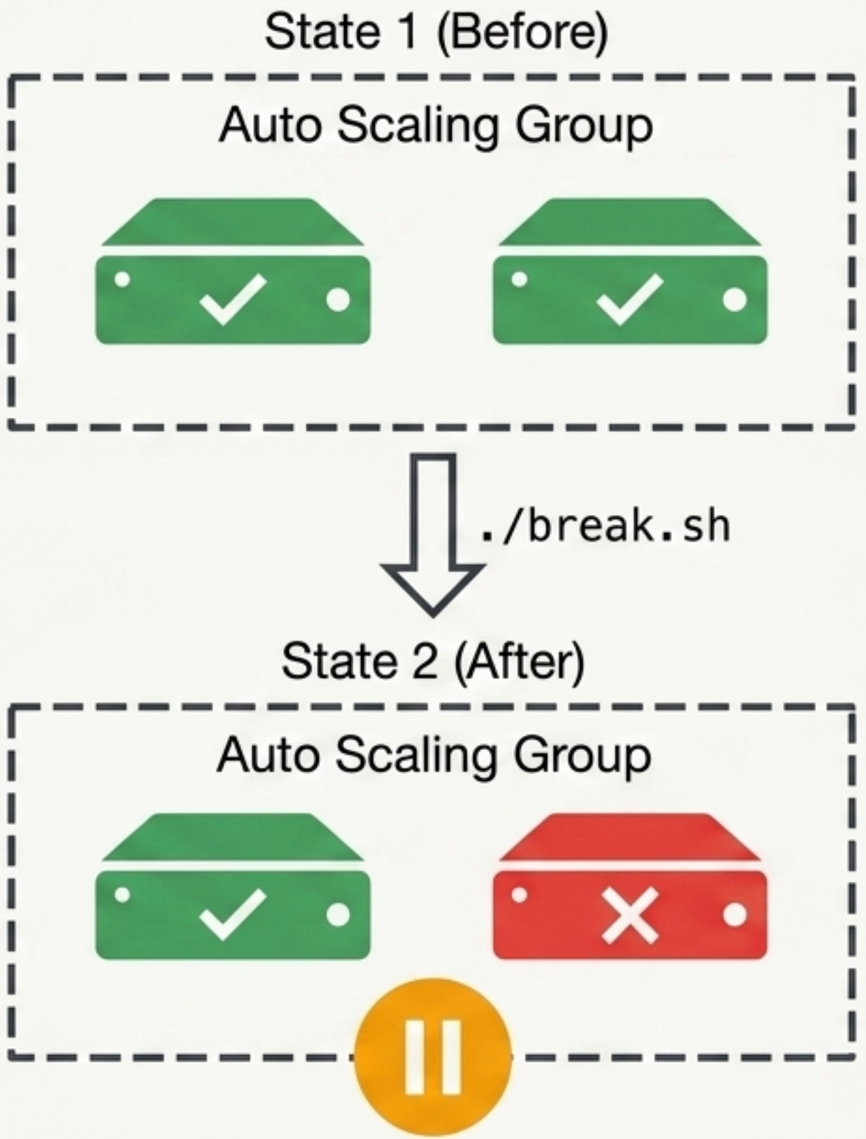
Step 1: Setup Environment

```
$ export ASG_NAME from terraform output
```

Step 2: Inject Failure

```
$ ./break.sh
```

Node Map



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Terminal Timeline

Continue from the previous editorial incident response runbook.

Step 3: Query the Control Plane

```
$ aws autoscaling
describe-auto-scaling-
groups
```

Annotated Output


```
{
  "AutoScalingGroups": [
    {
      "AutoScalingGroupName": "web-asg-1",
      "DesiredCapacity": 2,
      "MinSize": 1,
      "MaxSize": 5,
      "Instances": [
        {
          "InstanceId": "i-0a1b2c3d4e5f6g7h8",
          "LifecycleState": "InService"
        },
        {
          "InstanceId": "i-0a1b2c3d4e5f6g7h9",
          "LifecycleState": "Terminated"
        }
      ],
      "SuspendedProcesses": [
        "ReplaceUnhealthy"
      ],
    }
  ]
}
```


Diagnosis: The Suspended field is not empty. The automatic healing mechanism has been deliberately paused.

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Terminal Timeline

Continue from the previous slides

Step 4: Query the Information Plane

```
$ aws autoscaling
describe-scaling-activities
```

The Group’s Own Diary

```
[BLANK - No recent launch
activity recorded]
```

Instance terminated (Unhealthy).

Instance launched successfully.

The ASG logged the instance death, but recorded no subsequent launch activity. The control loop is broken.



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Terminal Timeline

Continue from the previous slides

Step 5: Reactivate Healing

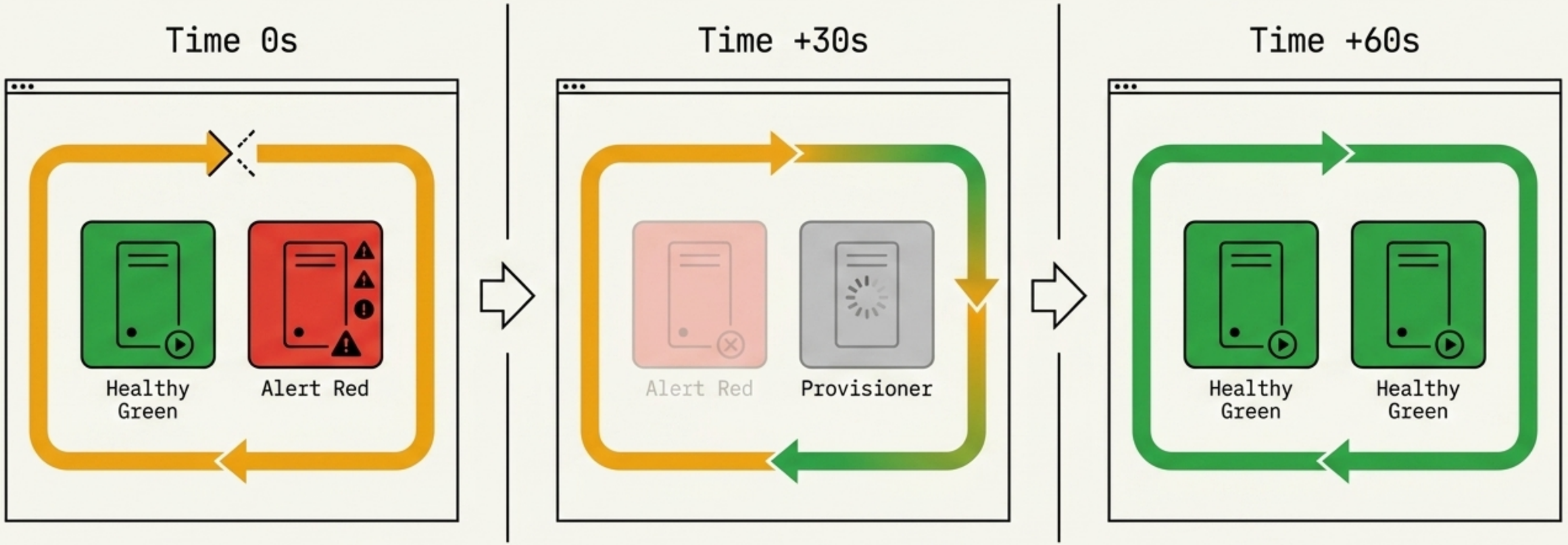
```
$ aws autoscaling resume-processes
```



Action: Turn healing back on.

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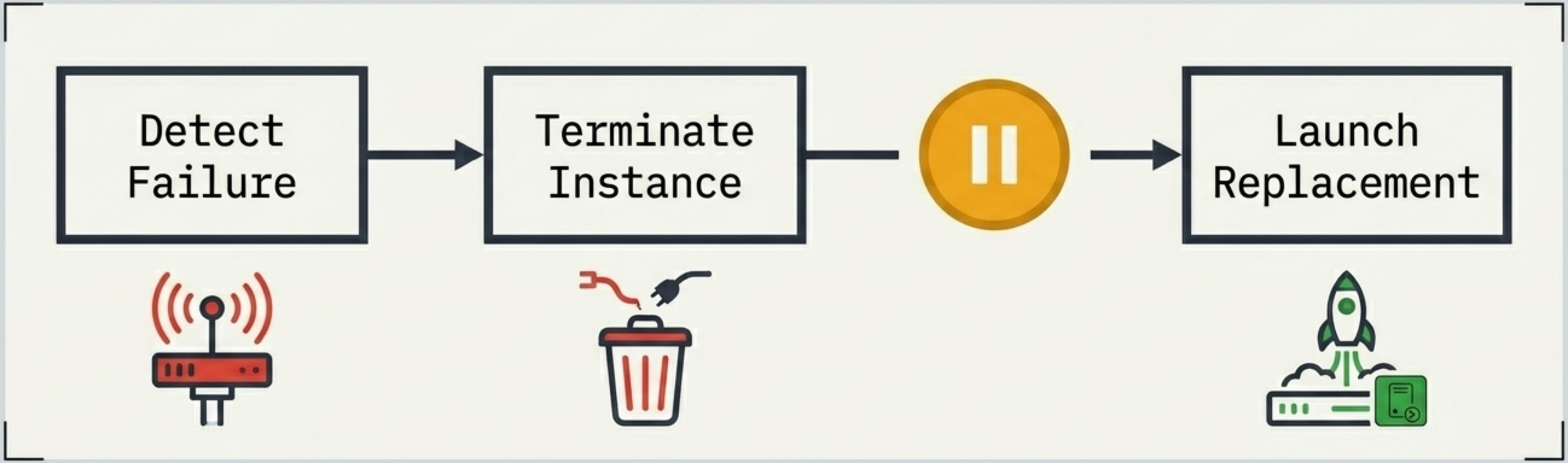
Step 6: System Recovery & Verification



Verify: Running instance count climbs back to 2 within approximately 60 seconds.

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The Suspended State Concept



Suspended processes silently pause execution. The ASG can still detect failures and terminate bad nodes, but the bridge to launching replacements is severed.

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The Auto Scaling Lifecycle

1. The Sensor (Health Check)

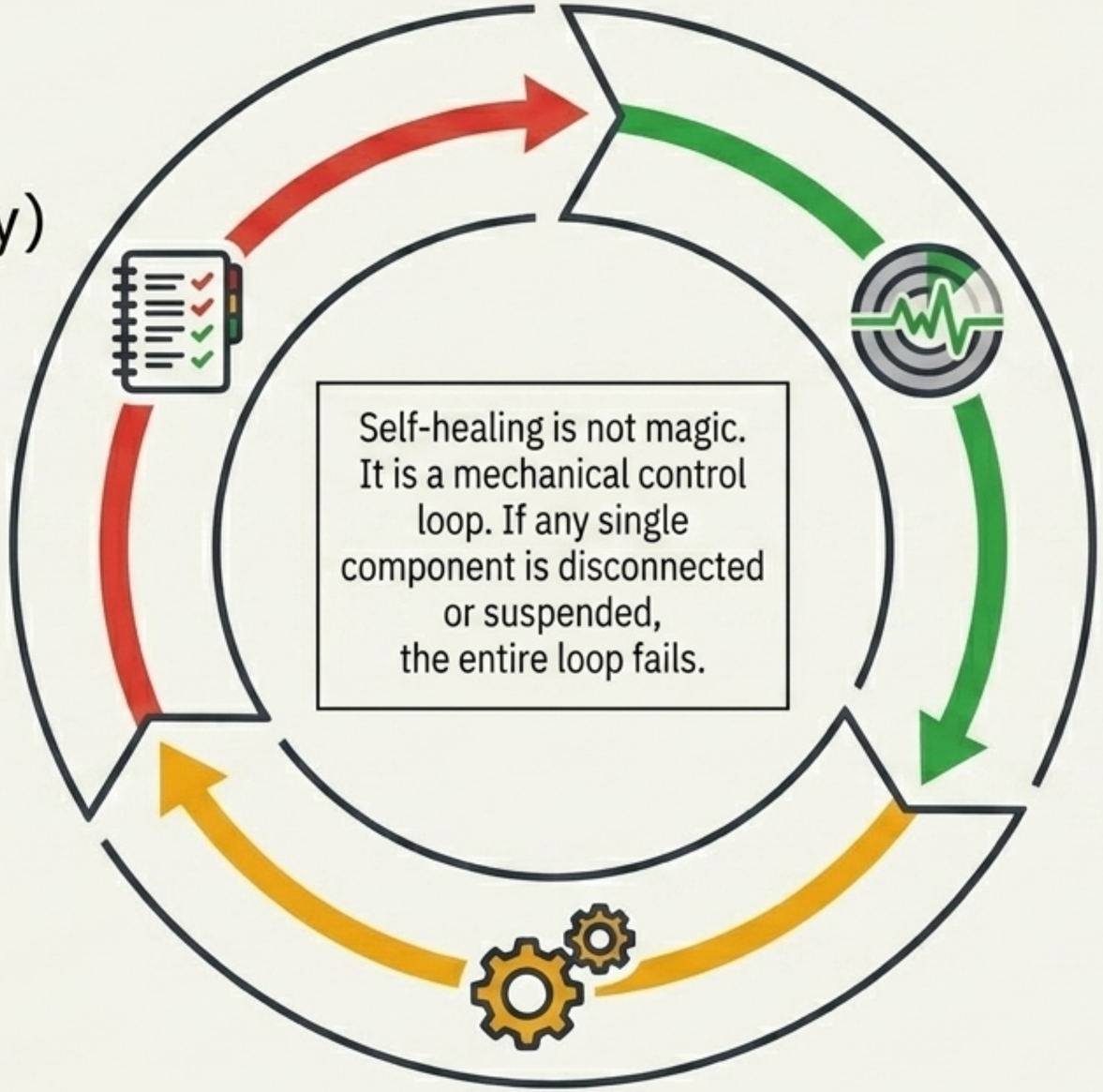
Detects state.

2. The Engine (ASG Processes)

Executes termination and replacement.

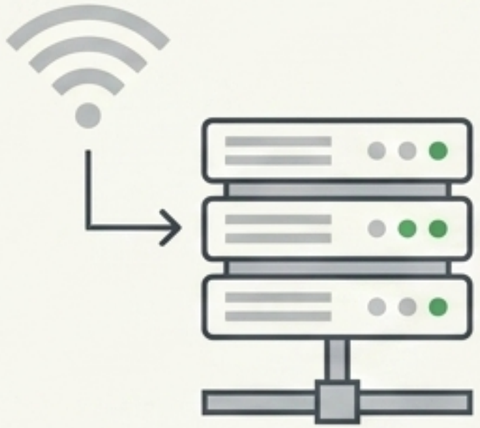
3. The Log (Activity History)

The group's own diary of actions taken.



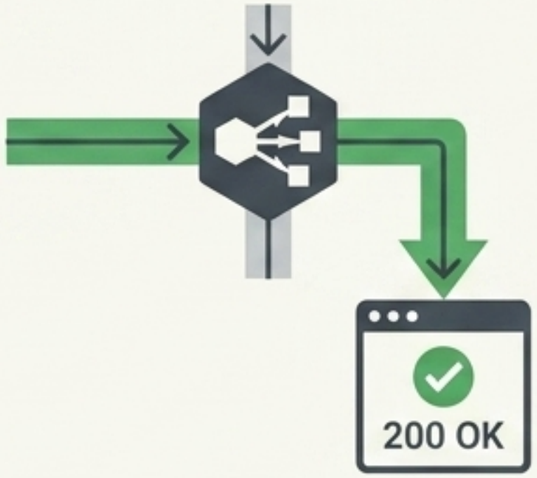
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The Sensor’s Role: Linking Opinions to Replacement



EC2 Health Check

Default EC2 checks only know if the underlying hardware is running.



`HealthCheckType=ELB`

Links the load balancer’s opinion to replacement. If the application stops responding to HTTP requests, the ASG engine is triggered to replace it.

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Diagnostic Matrix: The Dual Nature of ELB Health Checks

Scenario	Trigger	ASG Action	System Result
The Ideal (Healing)	True Instance Failure (Hardware crash).	ELB marks Unhealthy -> ASG terminates and replaces.	 System successfully heals.
The Flip Side (Death Loop)	Application overload or bad route (EC2 hardware is perfectly fine).	ELB marks Unhealthy -> ASG terminates -> New instance launches -> Overloads immediately.	 Infinite replacement loop. Healthy boxes are destroyed repeatedly.

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The Rule

An ASG only self-heals if its processes are running.

The Warning

A forgotten suspend is a time bomb — always resume it.

Incident Resolved. End of Runbook.

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